

Biosecurity research is part of ABARES' applied social research and analysis. Reports have been prepared for the Department of Agriculture Fisheries and Forestry, other government agencies, research and development corporations, and industry bodies.

Published: 26 March 2026

Overview

Vertebrate pests and weeds affect not only agricultural production and the environment, but also the wellbeing of individuals, households and communities. This report examines the social impacts of vertebrate pests and weeds, drawing on published literature and case study examples to identify how these impacts are experienced, how they vary by context, and which groups are most affected.

Using social impact assessment and a capitals framework, the report provides a structured way to identify and categorise the wide range of positive and negative social impacts associated with pests and weeds and their management. The findings support more socially informed policy and program design by improving understanding of how pest and weed pressures interact with human, community and environmental systems.

Key findings

- A diverse range of social impacts associated with vertebrate pests and weeds were identified in the literature, affecting individuals, households, businesses and communities through both direct impacts and indirect effects of management responses.
- Social impacts span multiple forms of capital, including human, social, political, financial, cultural, natural and built capital, and can be experienced as both negative and positive, depending on the pest or weed species, the context and stakeholder perspectives.
- The type and severity of social impacts vary by local context, with differences observed across regions, land uses and community settings, as illustrated through case study examples.
- Different groups experience impacts differently, with variation linked to roles, livelihoods, values and exposure to pests, weeds and management activities.
- Applying a capitals framework provides a systematic way to identify, compare and communicate social impacts, helping to reveal interactions, cumulative effects and trade-offs that may not be visible when impacts are considered in isolation.

Download

- [Applying social impact assessment and the capitals framework to understand and categorise social impacts Report - \(PDF 3.12 MB\)](#)
- [Applying social impact assessment and the capitals framework to understand and categorise social impacts Report - \(Word 4.9 MB\)](#)

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Published: 26 July 2023

Overview

Alien invasive species (AIS) cause a series of direct and indirect impacts and associated costs, on agricultural production, ecosystem function and the community. Understanding the total costs (market and non-market) of AIS is important for determining what level of investment delivers the best net benefits for society, and how to prioritise between competing management demands.

This paper provides environmental managers in government agencies with a guide on how best to consider non-market values when prioritising investments in the management of invasive species.

Key findings

- There is a clear case for systematically considering non-market impacts and, where possible, non-market values, in decision making regarding AIS.
- The likely consequence of overlooking non-market impacts is a socially sub-optimal level of investment into invasive species management in terms of the relative size of investment, and a misallocation of resources in terms of species targeted and choice of management actions.
- Methods for valuing non-market impacts are well developed and their implementation is feasible and can be incorporated as a monetary or non-monetary inputs into a benefit cost framework. However, significant institutional capacity building is likely to be required before these methods can be widely applied at low cost.

Download

[Methods for non-market valuation of alien invasive species - \(PDF 994 KB\)](#) [Methods for non-market valuation of alien invasive species - \(DOCX 4.24 MB\)](#) [Guide - Methods for non-market valuation of alien invasive species - \(PDF 217 KB\)](#) [Guide - Methods for non-market valuation of alien invasive species - \(DOCX 703 KB\)](#)

If you have difficulty accessing these files, visit [web accessibility](#) for assistance.

ABARES is developing a manual for Program Design, Monitoring and Evaluation of General Surveillance Programs. It will provide a step-by-step guide to support general surveillance program coordinators and teams to design general surveillance programs, including a monitoring and evaluation component.

More on the [monitoring and evaluation manual](#)

Summary

Published: 19 April 2022

Engaging people from all walks of life in the monitoring and reporting of pests, weeds and diseases is called general surveillance. Instigating and maintaining general surveillance programs are no easy feat. The General surveillance program guidelines (the Guidelines) have been developed to help program coordinators, policy-makers, funders and those who monitor and evaluate general surveillance

programs to understand the key considerations for designing, planning and implementing such programs. It is relevant to all biosecurity sectors, including plant, animal, marine, weed and the environment.

The Guidelines are based on systems thinking and the lessons learned from 9 case studies from across plant, animal, marine, weeds and the environment sectors in Australia and New Zealand. The Guidelines have been refined with feedback from 140 experts who work on different parts of general surveillance programs.

Published: 20 August 2019

Overview

Marine pests can cause significant negative social, ecological and economic impacts to infrastructure, marine habitats, water quality, marine industries and coastal amenity values. Maintaining an effective marine pest biosecurity system that minimises the risk of marine pests to Australia is a priority for the Australian Government. The Department of Agriculture commissioned ABARES to investigate the current state of Australia's marine pest biosecurity stakeholder network by means of a social network analysis.

Key Issues

The findings of the study provide a broad understanding of the current marine pest stakeholder network by identifying key players in the network and relationships, and patterns of interaction, between them. The study showed that involvement and interest in marine pest biosecurity is extensive and complex. A wide range of government and non-government organisations and groups participate in the network. The analysis identified opportunities to tap into existing stakeholder networks and build on current structures to further improve network function.

Published: 24 July 2018

Overview

This report presents key results from a national survey of 1,585 recreational boat operators about their vessel biofouling management actions. The purpose was to inform a national communication approach to reducing the risk of marine pest translocation via biofouling of recreational boats in the Australian marine environment.

Biofouling is a term for the growth on a boat's hull that can move invasive pests and diseases from one marine environment to another, which is a biosecurity risk.

Australia's national approach to domestic marine pest biosecurity relies on voluntary uptake of the national biofouling management guidelines by recreational vessel operators, to prevent and manage biofouling growth.

Key Issues

The study found that a large proportion of recreational boat operators already undertake a range of effective biofouling management actions on a regular basis, with more than 60 per cent using best-practice approaches, such as regularly cleaning the boat hull and niche areas of the boat, renewing anti-fouling coatings each year and capturing biofouling waste after cleaning.

The majority of respondents demonstrated considerable interest in doing the right thing to protect the environment. The analysis presents opportunities to develop messaging that could be used as part of future engagement strategies with recreational boaters to influence behaviour voluntarily and promote best practice biofouling management activities.

Published: 2 May 2017

The Pest animal and Weed Management Survey: National landholder survey results report presents the key results from a national survey of 6470 agricultural land managers undertaken by ABARES in 2016 about pest and weed management on their property and local area.

The survey respondents represented land managers across broadacre, horticulture, dairy and other livestock (poultry, deer, horses, bee-keeping) industries, each with an estimated value of agricultural operations (EVAO) of \$5000 per year or more, across 53 natural resource management regions in Australia.

The data were collected through a combination of hardcopy postal and online versions of the survey. Approximately 77 per cent of responses received were via the postal survey and 23 per cent via the online survey. A response rate of 50.1 per cent overall was achieved.

This report presents results on a range of topics from the survey including:

- level of awareness of pest animals and Weeds of National Significance (WoNS)
- impacts of pest animals and weeds
- pest animal and weed management activities on the property and in the local area
- information sources and participation in local support networks.

Published: 27 July 2015

Wild dog management 2010 to 2014: National landholder survey results presents results and analysis from a national survey of Australian sheep and cattle landholders in late 2014, in areas affected by wild dogs. The results are combined with data from a similar 2010 survey to assess longitudinal changes in wild dog impacts and management activities. Around 1,010 landholders participated in the 2014 survey.

The survey examined landholders' perspectives on wild dog problem severity and distribution, personal and economic impacts and collective management actions. The study also quantifies factors that influence the effectiveness of wild dog management groups and their achievement of outcomes.

Published: 24 April 2014

An integrated assessment of the impact of wild dogs in Australia evaluates the economic, environmental and social impacts of wild dogs in Australia. It assesses the costs and benefits of investing in wild dog management to prioritise future investments using a cost-benefit analysis framework. Integrating the economic impacts of wild dogs on Australian agriculture with non-market environmental and social impacts enabled a more accurate estimation of the return to the entire Australian community of investments to control wild dogs.

Published: 29 October 2013

This review forms part of a larger Australian Wool Innovation-funded project on a landscape-scale approach to managing wild dogs in Australia. Specifically, it covers existing research on the social impacts of wild dog attacks on livestock, including social and psychological impacts on individual farmers and livestock enterprises.

Based on this research, it identifies major sources of tension between stakeholders, and discusses possible collaborative approaches to managing wild dogs that may help resolve these tensions.

Published: 27 July 2015

Participatory wild dog management: Views and practices of Australian wild dog management groups examines the features of Australian wild dog management groups, particularly in terms of landholder participation and collaboration, to identify what helps or hinders the groups in achieving coordinated and effective wild dog management. The study was based on interviews with representatives from 30 groups across Australia.

Published: 11 October 2013

Australia is free of Foot and Mouth Disease (FMD), a highly contagious disease affecting cloven-hoofed animals including cattle, sheep, goats, pigs, deer and camels. International experience shows an outbreak of FMD results in trade bans on livestock products to export destinations due to the risk of transmission of the disease to livestock. For large exporters, this results in product being diverted to the domestic markets, as the meat is safe for human consumption, reducing product prices.

This report shows an outbreak of FMD in Australia is expected to generate very large adverse economic impacts to both producers and other industries inside and beyond the outbreak area; with financial losses and eradication activities also having social impacts. The revenue losses to producers are larger than previously estimated by the Productivity Commission (2002). The higher figures in this study are a result of additional expected market access requirements from trading partners which in turn results in a longer expected time out of the market and a greater loss of market share.

Findings suggest that these economic and social impacts can be reduced by the choice of eradication strategy - with results indicating that vaccination could play a beneficial role where spread is rapid in high density production areas. Impacts can also be reduced by resuming market access quickly where feasible, improving response preparedness (though surveillance, eradication arrangements and livestock tracing) and the use of communication and support before and during an outbreak.

Publication date: 3 October 2007

This report explores the land uses, land management practices and motivations of small landholders in these regions.

The research focused on identifying practices that may give rise to exotic pests and diseases currently not established in Australia.

Small landholders in peri-urban areas are highly diverse and partake in a wide range of land uses; however they largely fall into two categories: 'lifestylers' and 'farmers'.

The report provides recommendations for understanding and communicating with these landholders, as well as recommendations for future research.

Publication date: 30 January 2006

This study focuses on peri-urban and rural lifestyle landowners as a possible biosecurity risk group, and reports on literature findings and case studies conducted in Western Australia, Victoria and Queensland.

Source: <https://www.agriculture.gov.au/abares/research-topics/social-sciences/biosecurity>